**D 122513** **(Pages : 4)** **Name.....**

**Reg. No.....**

**SECOND SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, APRIL 2025**

(CBCSS)

Mathematics

MTH 2C 07—REAL ANALYSIS—II

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

**Part A**

*Answer* **all** *questions.*
*Each question has weightage* 1.

1. Verify whether $\{1, 2, 3\}$ is a Borel set.

2. Let $E_n = \left(-\frac{1}{n}, 1+\frac{1}{n}\right)$. Find $m\left(\bigcap_n E_n\right)$ where $m$ is the Lebesgue measure.

3. Let

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{otherwise} \end{cases}.$$

Verify whether $f$ is a Lebesgue measurable function on $\mathbb{R}$.

4. Let

$$f(x) = \begin{cases} 1 & \text{if } 0 \le x < 1 \\ 0 & \text{if } 1 \le x \le 2 \end{cases}.$$

Verify whether $f(x)$ is a characteristic function.

**Turn over**

5. Let $f(x)$ in $[0,1]$ be defined by

$$f(x) = \begin{cases} 0 \text{ if } x \text{ is rational} \\ 1 \text{ otherwise} \end{cases}.$$

Find $\int_E f$ where E is the set of all irrationals in $[0,1]$.

6. Consider the family of functions

$$F = \left\{ \chi_{\left[-\frac{1}{n}, \frac{1}{n}\right]} : n \in \mathbb{N} \right\}.$$

Verify whether F is uniformly integrable in $[-1,1]$.

7. Let $(q_n)_{n=1}^{\infty}$ be an enumeration of the rationals in $(0,1)$ and let

$$f(x) = \sum_{n : q_n \le x} \frac{1}{2^n}$$

for $0 < x < 1$. Show that $f$ is not continuous at $q_k$ for any $k$.

8. Let $f(x)$ be continuous and increasing on $[0.1]$ Show that $f(x)$ is absolutely continuous on $[0,1]$.

(8 × 1 = 8 weightage)

## Part B

*Answer any* **two** *questions from each module.*
*Each question has weightage* 2.

MODULE I

9. Let C be a countable subset of $\mathbb{R}$. Show that $m^*(C) = 0$ where $m^*$ is the Lebesgue outer measure on $\mathbb{R}$.

10. Show that if $m^*(E) = 0$ then E is Lebesgue measurable.

11. Show that if $(A_n)$ is an ascending chain of Lebesgue measurable sets then

$$m\left(\bigcup_n A_n = \lim_{n\to\infty} m(A_n)\right).$$

MODULE II

12. Let E be a measurable set of finite measure and $\phi$, $\psi$ be simple functions on E such that $\phi \le \psi$. Show that

$$\int_E \phi \le \int_E \psi.$$

13. Let E be a set of measure zero and $f$ be a bounded measurable function on E. Show that $\int_E f = 0$.

14. Let $(f_n)$ be a sequence of non negative integrable functions on a measurable set E and let $\lim_{n\to\infty} \int_E f_n = 0$. Show that $(f_n)$ converges to 0 in measure on E.

MODULE III

15. Let $f$ be a function on $[0,1]$ such that $f(x) = g(x) - h(x)$ for all $x \in [0,1]$ where $g$ and $h$ are increasing functions. Show that $f$ is of bounded variation on [0,1].

16. Let $f$ be absolutely continuous on $[0,1]$. Show that the total variation function $\mathrm{TV}(f)$ is also absolutely continuous on $[0,1]$.

17. Let $\phi$ be a convex function on a bounded interval $(a,b)$. Show that $\phi$ is a Lipschitz function on every subinterval $[c,d]$ of $(a,b)$.

(6 × 2 = 12 weightage)

**Turn over**

## Part C

*Answer any* **two** *questions.*
*Each question has weightage* 5.

18. (a) Define Lebesgue measurable set.

    (b) Show that union of a finite collection of measurable sets is measurable.

    (c) Show that the union of a countable collection of measurable sets is measurable.

19. Let $f, g$ be measurable functions on a measurable set E. Show that

    (a) $f+g$ is measurable.

    (b) $f^2$ is measurable.

    (c) if $h(x)=f(x)\,g(x)$ for all $x \in \mathrm{E}$ then $h$ is measurable.

20. (a) Define convergence in measure.

    (b) Let $(f_n)$ be a sequence of functions defined on a measurable set E of finite measure. Suppose that $f_n \to f$ pointwise a.e. on E and that $f$ is finite a.e. on E. Show that $f_n \to f$ in measure on E.

21. (a) Define indefinite integral of a function.

    (b) Show that a function $f$ on a closed and bounded interval $[a, b]$ is an indefinite integral over $[a, b]$ if and only if $f$ is absolutely continuous on $[a, b]$.

(2 × 5 = 10 weightage)